

Answer: C

25 Let the length of each grid be R .

$$E_1 = \frac{Q}{4\pi\epsilon_0 R^2} = 32.0 \text{ N C}^{-1}$$

$$QR = \sqrt{(2R)^2 + R^2} = \sqrt{5}R$$

$$E_R = \frac{Q}{4\pi\epsilon_0 (\sqrt{5}R)^2} = \frac{32.0}{5} = 6.4 \text{ N C}^{-1}$$

$$E_s = \frac{Q}{4\pi\epsilon_0 (3R)^2} = \frac{32.0}{9} = 3.6 \text{ N C}^{-1}$$

Answer: C

26 When the separation of the parallel plates is increased, the electric field between the plates